

Tutorial 3

Questions

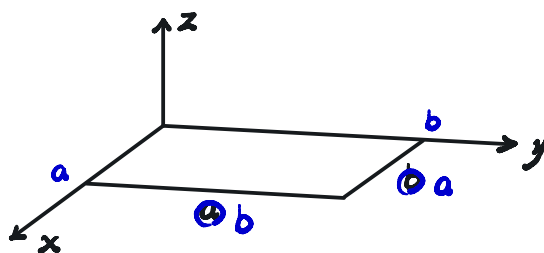
Consider a rectangular plate with clamped unloaded edges of length a and simply supported loaded edges of width b ($b < a$). A uniform compressive membrane force N_x acts on the loaded edges. The approximated transversal deformation of the buckled plate is described by the following family of shape functions (i.e. buckling modes)

$$w = A_{mn} \sin \frac{m\pi x}{a} \left(1 - \cos \frac{2n\pi y}{b} \right); \quad m, n \in \mathbb{N}^+$$

- 1) Explain why the shape function given above is suitable for predicting the buckling of the plate via the Rayleigh-Ritz method.
[10 marks]
- 2) Calculate the strain energy of the plate for the load and assumed deflection stated above.
[30 marks]
- 3) Calculate the external work performed by the in-plane membrane force for N_x the buckling modes given above.
[20 marks]
- 4) Find the expression of the buckling load of the plate as a function of the aspect ratio a/b .
[20 marks]
- 5) Compute the aspect ratio of the plate a/b that yields the minimum buckling load, as well as the value of the latter.
[20 marks]

Some useful integrals:

$$\int_0^a \sin^2 \frac{m\pi x}{a} dx = \int_0^a \cos^2 \frac{m\pi x}{a} dx = \frac{a}{2}, \quad m \in \mathbb{N}^+$$



1) for Rayleigh-Ritz Method, essential (geometry) boundary condition should be satisfied

$$\therefore w(x, 0) = w(x, b) = 0 \text{ for } 0 \leq x \leq a$$

$$w(0, y) = w(a, y) = 0 \text{ for } 0 \leq y \leq b$$

$$\text{and } \boxed{\frac{dw}{dx}(x, 0) = \frac{dw}{dx}(x, b) = \frac{dw}{dy}(x, 0) = \frac{dw}{dy}(x, b) = 0 \text{ for } 0 \leq x \leq a}$$

$$\therefore w = A_{mn} \sin \frac{m\pi x}{a} (1 - \cos \frac{2n\pi y}{b}),$$

$$\therefore \frac{dw}{dx} = A_{mn} \frac{m\pi}{a} \cos \frac{m\pi x}{a} (1 - \cos \frac{2n\pi y}{b}),$$

$$\frac{dw}{dy} = A_{mn} \frac{2n\pi}{b} \sin \frac{m\pi x}{a} \sin \frac{2n\pi y}{b}$$

$\therefore$ satisfied boundary condition listed above

$$\frac{dw}{dx}(0, y) \neq 0, \frac{dw}{dx}(a, y) \neq 0 \text{ for } 0 \leq y \leq b$$

* $\frac{dw}{dy}$ rotation along the edge parallel to the x-axis

$\frac{dw}{dx}$ rotation along the edge parallel to the y-axis

$$2) U = \frac{1}{2} \iiint \sigma \epsilon \, dV$$

$$= \frac{1}{2} \int_0^a \int_0^b \int_0^h \sigma_{xx} z \, dz \frac{\epsilon_{xx}}{z} + \int_0^a \int_0^b \int_0^h \sigma_{yy} z \, dz \frac{\epsilon_{yy}}{z} + \int_0^a \int_0^b \int_0^h \tau_{xy} z \, dz \frac{\gamma_{xy}}{z} \, dx \, dy$$

$$= \frac{1}{2} \int_0^a \int_0^b M_x \frac{\epsilon_{xx}}{z} + M_y \frac{\epsilon_{yy}}{z} + M_{xy} \frac{\gamma_{xy}}{z} \, dx \, dy$$

$$= \frac{1}{2} D \int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \frac{\partial^2 w}{\partial x^2} + \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \frac{\partial^2 w}{\partial y^2} + (1-\nu) \frac{\partial^2 w}{\partial x \partial y} \left(2 \frac{\partial^2 w}{\partial x \partial y} \right) \, dx \, dy$$

$$= \frac{1}{2} D \int_0^a \int_0^b \left(\left(\frac{\partial^2 w}{\partial x^2} \right)^2 + \left(\frac{\partial^2 w}{\partial y^2} \right)^2 + 2\nu \left(\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \right) + 2(1-\nu) \left(\frac{\partial^2 w}{\partial x \partial y} \right)^2 \right) \, dx \, dy$$

$$\text{where } D = \frac{Et^3}{12(1-\nu^2)}$$

$$\frac{\partial^2 w}{\partial x^2} = -A_{mn} \left(\frac{m\pi}{a} \right)^2 \sin \frac{m\pi x}{a} (1 - \cos \frac{2n\pi y}{b}),$$

$$\frac{\partial^2 w}{\partial y^2} = A_{mn} \left(\frac{2n\pi}{b} \right)^2 \sin \frac{m\pi x}{a} \cos \frac{2n\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x \partial y} = A_{mn} \left(\frac{m\pi}{a} \right) \left(\frac{2n\pi}{b} \right) \cos \frac{m\pi x}{a} \sin \frac{2n\pi y}{b}$$

$$\therefore \left(\frac{\partial^2 w}{\partial x^2} \right)^2 = A_{mn}^2 \left(\frac{m\pi}{a} \right)^4 \sin^2 \frac{m\pi x}{a} (1 - 2\cos \frac{2n\pi y}{b} + \cos^2 \frac{2n\pi y}{b})$$

$$\left(\frac{\partial^2 w}{\partial y^2}\right)^2 = A_{mn}^2 \left(\frac{m\pi}{b}\right)^4 \sin^2 \frac{m\pi x}{a} \cos^2 \frac{2n\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} = -A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \left(\frac{2n\pi}{b}\right)^2 \sin^2 \frac{m\pi x}{a} \left(\cos \frac{2n\pi y}{b} - \cos^2 \frac{2n\pi y}{b}\right)$$

$$\left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \left(\frac{2n\pi}{b}\right)^2 \cos^2 \frac{m\pi x}{a} \sin^2 \frac{2n\pi y}{b}$$

$$\therefore \int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x^2}\right)^2 dx dy = A_{mn}^2 \left(\frac{m\pi}{a}\right)^4 \left(\frac{a}{2}\right) \left(b - 0 + \frac{b}{2}\right) = A_{mn}^2 \frac{3m^4 \pi^4 b}{4a^3}$$

$$\int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial y^2}\right)^2 dx dy = A_{mn}^2 \left(\frac{2n\pi}{b}\right)^4 \left(\frac{a}{2}\right) \left(\frac{b}{2}\right) = A_{mn}^2 \frac{4n^4 \pi^4 a}{b^3}$$

$$\int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2}\right) dx dy = -A_{mn}^2 \left(\frac{4m^2 n^2 \pi^4}{a^3 b^3}\right) \left(\frac{a}{2}\right) \left(0 - \frac{b}{2}\right) = A_{mn}^2 \frac{m^2 n^2 \pi^4}{ab}$$

$$\int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 dx dy = A_{mn}^2 \left(\frac{4m^2 n^2 \pi^4}{a^3 b^3}\right) \left(\frac{a}{2}\right) \left(\frac{b}{2}\right) = A_{mn}^2 \frac{m^2 n^2 \pi^4}{ab}$$

$$\begin{aligned} \therefore U &= \frac{D}{2} \left[A_{mn}^2 \frac{3m^4 \pi^4 b}{4a^3} + A_{mn}^2 \frac{4n^4 \pi^4 a}{b^3} + 2\nu A_{mn}^2 \frac{m^2 n^2 \pi^4}{ab} + 2(1-\nu) A_{mn}^2 \frac{m^2 n^2 \pi^4}{ab} \right] \\ &= \frac{1}{8} A_{mn}^2 D \pi^4 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2 n^2}{ab} \right) \end{aligned}$$

$$3) W_{ip} = \frac{1}{2} \int_0^a \int_0^b N_x \left(\frac{dw}{dx}\right)^2 dx dy$$

$$= \frac{N_x}{2} \int_0^a \int_0^b A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \cos^2 \frac{m\pi x}{a} \left(1 - 2\cos \frac{2n\pi y}{b} + \cos^2 \frac{2n\pi y}{b}\right) dx dy$$

$$= \frac{N_x}{2} A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \left(\frac{a}{2}\right) \left(b - 0 + \frac{b}{2}\right)$$

$$= \frac{1}{8} N_x A_{mn}^2 \pi^2 \left(\frac{3m^2 b}{a}\right)$$

$$4) \pi = U - W = U - (W_{op} + W_{ip})$$

$$= \frac{1}{8} A_{mn}^2 \pi^4 \left[D \pi^2 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2 n^2}{ab} \right) - N_x \left(\frac{3m^2 b}{a} \right) \right]$$

$$\therefore \frac{d\pi}{dA_{mn}} = \frac{1}{4} A_{mn} \pi^4 \left[D \pi^2 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2 n^2}{ab} \right) - N_x \left(\frac{3m^2 b}{a} \right) \right]$$

$$\text{At the buckling load, } N_{xs}, \frac{d\pi}{dA_{mn}} = 0$$

$$\therefore D \pi^2 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2 n^2}{ab} \right) = N_{xs} \left(\frac{3m^2 b}{a} \right)$$

$$\therefore N_{xs} = D \pi^2 \left(\frac{m^2}{a^2} + \frac{16n^4 a^2}{3m^2 b^3} + \frac{8n^2}{3b^2} \right) = \frac{D \pi^2}{b^2} \left[m^2 \left(\frac{a}{b} \right)^2 + \frac{16n^4}{3m^2} \left(\frac{a}{b} \right)^2 + \frac{8n^2}{3} \right]$$

$$5) \frac{dN_{x8}}{d(a/b)} = \frac{Dx^*}{b^2} \left[-2m^* \left(\frac{a}{b} \right)^{-3} + \frac{32n^*}{3m^*} \left(\frac{a}{b} \right) \right]$$

To find minimum buckling load, $\frac{dN_{x8}}{d(a/b)} = 0$

$$\therefore -2m^* \left(\frac{a}{b} \right)^{-3} + \frac{32n^*}{3m^*} \left(\frac{a}{b} \right) = 0$$

for $\frac{a}{b} \neq 0$ and $\frac{a}{b} > 0$.

$$-6m^* + 32n^* \left(\frac{a}{b} \right)^4 = 0$$

$$\therefore \frac{a}{b} = \frac{3^{\frac{1}{4}} m}{2n}$$

$$\therefore (N_{x8})_{min} = \frac{Dx^*}{b^2} \left(m^* \frac{4n^*}{\sqrt{3}m^*} + \frac{16n^*}{3m^*} \frac{\sqrt{3}m^*}{4n^*} + \frac{8n^*}{3} \right)$$

$$= \frac{Dx^*}{b^2} \left(\frac{4\sqrt{3}}{3} n^* + \frac{4\sqrt{3}}{3} n^* + \frac{8n^*}{3} \right)$$

$$= \frac{8+8\sqrt{3}}{3} \frac{Dx^* n^*}{b^2}$$